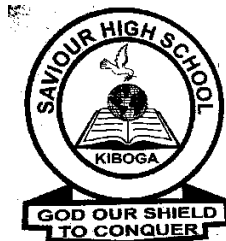


535/1
PHYSICS
Paper 1
(theory)
Oct/Nov. 2025
2 ¼ hours



SAVIOUR HIGH SCHOOL KIBOGA

S.1 End of term III examinations

PHYSICS

Paper 1

(theory)

2 hours 15 minutes

Instructions to the candidates

This paper consists of five examination items

*You are required to select **four** items in all*

Any additional item(s) responded shall not be scored

In your presentations, make clear drawings, neat hand writing and clear substitutions

Non programmable calculators and tables may be used.

At the end of the paper, fasten all your work together

Turn over

Item 1

The S.1 learners went to the physics laboratory for a lesson. The teacher had prepared the laboratory apparatus and materials for the learners to carry out an experiment in groups. However, the learners entered while running trying to get stools and ended up colliding with one another and breaking some of the apparatus.

In addition, the apparatus prepared were materials like a small irregular stone, water and a small regular wooden block of mass 800g and dimensions 15.0cm x 10.0cm x 6.0cm

Hint (the density of water = 1gcm^{-3})

Task

- (a) Guide the S.1 learners on how they should conduct themselves on entering the laboratory and while in the laboratory.
- (b) Describe how you would carry out the scientific investigation to determine the density of the stone.
- (c) (i) Calculate the density of the wooden block provided in kgm^{-3} and express your answer in scientific notation.
(ii) Explain what would be observed if the wooden block in (c)(i) above was immersed in water.

Item 2

A maid arrived home late in the night. She wanted to prepare the cup of tea in the kitchen. Since it was dark, she lit the candle and then lit the charcoal stove and placed there a saucepan containing water and covered it with the lid. A boy who had never prepared tea was with her in the kitchen and was observing all that was going on. To his surprise, the size of the candle was reducing and the top part was accumulating the liquid, though the candle kept lighting. When water boiled, she removed the lid from the saucepan, steam poured out and he was again surprised to see water droplets formed on the lid.

She poured hot water in the cup and later added tea leaves in the cup. All the water turned brown even before stirring. All these observations puzzled the boy more.

Task

- (a) Identify the states of matter in the scenario and give examples.
- (b) Using the kinetic theory of matter, explain the arrangement of particles in the state(s) of matter identified in (a) above.
- (c) Name the change(s) of the states of matter in the scenario.
- (d) Explain why all the water in the cup turned brown when the tea leaves were added in hot water.

Item 3

A man was driving a car on a marram road on a rainy day. The car kept on sliding on the road surface until it got stuck in the mud. A breakdown car was hired to pull the car out of the mud and the truck applied the force of 5600N and the car could not move out. A group of men had to help the truck by pushing the car from behind by applying the force of 3400N in the same direction as that of the breakdown truck and the car was moved out of the mud.

The rain increased and the men took shelter in the nearby house. They were surprised to see the wall of the house wet from the floor upwards and the paint as peeling off. Even on the surface of stagnant water in the compound after the rain had stopped, were insects walking. The men got confused.

Task

As a learner of physics;

- (a) Explain what could have caused the sliding of the car on the road surface and how this could be avoided.
- (b) Explain what could have caused the peeling off of the paint of the wall in the house and how it can be avoided.
- (c) Explain why the insects were able to float and walk on the water surface.
- (d) What is the total force that was used to pull the car?

Item 4

A boy was not feeling well so he visited the clinic for checkup. A nurse wanted to take the temperature of the boy using the digital thermometer but the nurse discovered that it did not have dry cells in it. The nurse opted to use an uncalibrated thermometer. When the uncalibrated thermometer was immersed in pure melting ice, the mercury thread was 3.0cm long. When immersed in steam, the mercury thread rose to 25.0cm and when the thermometer was put under the patient's armpits, the length of the mercury thread became 11.5cm.

The boy was asked to remove his shoes and stockings so as to step on the weighing scale for measurement of his mass. As he stepped on the tiled floor, he felt that it was cold. As he waited for the laboratory test results, he sat out in the sunshine since he was feeling cold. The boy was wearing a black shirt with the white trouser. He felt the upper part of his body was hotter than the lower part.

Task

- (a) (i) Help the nurse to obtain the value of the body temperature of the patient in the kelvin scale.
(ii) Using the value obtained (a) (i) above, comment on the health of the patient.
- (b) Explain why the tiled floor felt cold when the patient stepped on it.
- (c) Explain why the upper part of the patient's body felt hotter than the lower part.

Item 5

On Monday 08/04/2024, it turned dark during the day and many people in one village were confused of what happened. In order to view what was happening in the sky, some used a pin hole camera and others put water in the basin to view the sky.

Task

- (a) Using an appropriate diagram, explain to the people in village what happened on that day.
- (b) (i) Using a ray diagram, describe the characteristics of the images formed by the plane mirror.
(ii) How many images would be formed if an object was placed between two plane mirrors inclined at an angle of 60° to each other?
- (c) What is the size of the pin hole camera if an object is placed 18.0cm from a pin-hole forms an image which is 3 times the size of an object.

END

“wishing you success, Merry Christmas and Happy new year”